

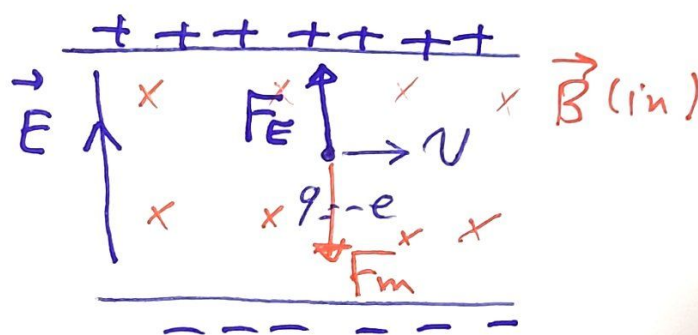
Set #4 - Solution - PHY102A MKQ

(19.3) NO, Because if the electron (charge) is subject to electric and magnetic forces act in opposite direction and of equal magnitude the net force on the electron (charge) is Zero

$$|\vec{F}_E| = |\vec{F}_m|$$

$$\vec{F}_E + \vec{F}_m = 0$$

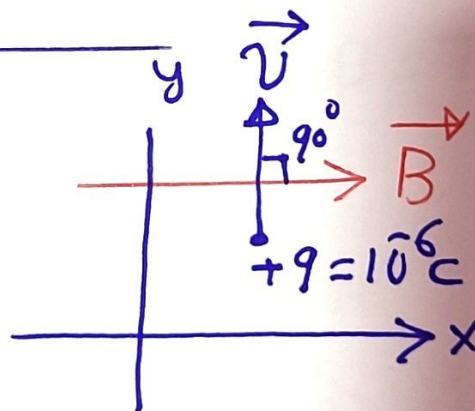
$$qE - qvB = 0$$



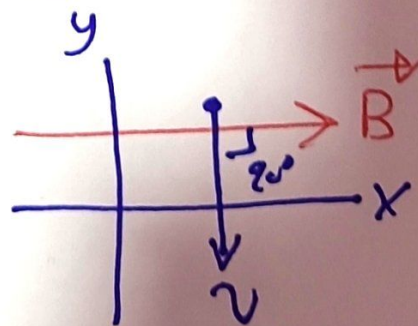
$$\text{(19.4)} \quad \vec{F}_m = q \vec{v} \times \vec{B}$$

$$(a) \quad F_m = qvB \sin(90^\circ)$$

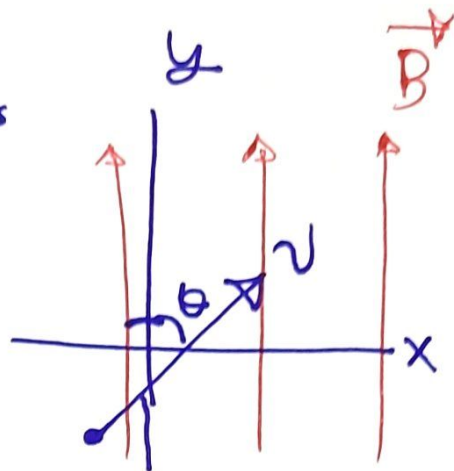
$$= 10^{-6} \times 10^4 \times 2 \times 1 = 2 \times 10^{-2} \text{ N (in or -ve z-axis)}$$



(b) $F_m = 2 \times 10^{-2} \text{ N}$ same as in part (a) but the direction of the force is (out) or +ve z-axis.



(119.7) Given $m = 0.01 \text{ kg}$, $v = 100 \text{ m/s}$
 $\theta = 30^\circ$, $B = 10^{-2} \text{ T}$, $q = -10^{-3} \text{ C}$

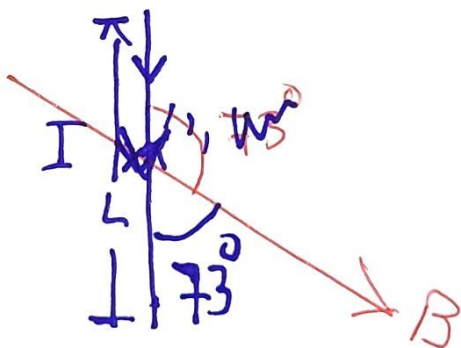


(a) $F_m = qvB \sin \theta$
 $= 10^{-3} \times 10^2 \times 10^{-2} \sin 30^\circ$
 $= 5 \times 10^{-4} \text{ N}$ direction into the page (-ve z-axis)
 because the charge is negative

(b) Newton's 2nd Law $F = ma = F_m$
 $ma = F_m \Rightarrow a = \frac{F_m}{m} = \frac{5 \times 10^{-4}}{0.01} = 5 \times 10^{-2} \text{ m/s}^2$

(119.12) Given $I = 10^3 \text{ A}$, $B = 7 \times 10^{-5} \text{ T}$, $\theta = 73^\circ$

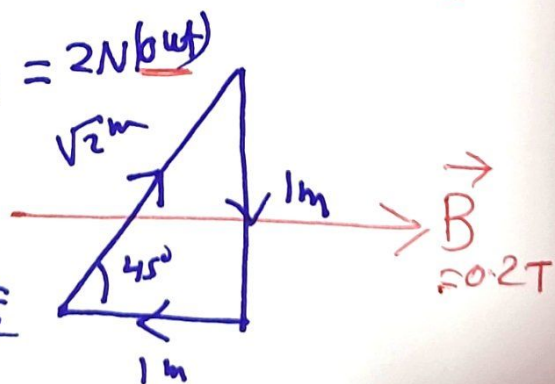
$F = ILB \sin \theta$
 $= 10^3 \times 10^2 \times 7 \times 10^{-5} \sin(73^\circ)$
 $= 6.7 \text{ N}$



(119.13) Right $F_m(\text{left}) = ILB \sin 90^\circ = 10 \times 1 \times 2 \times 1 = 2 \text{ N (left)}$

$F_m(\text{anti-parallel}) = 0$ bc $\sin 180^\circ = 0$

$F(\text{diagonal}) = ILB \sin 45^\circ = 10 \times \sqrt{2} \times 2 \times \frac{1}{\sqrt{2}}$
 $= 2 \text{ N (in)}$



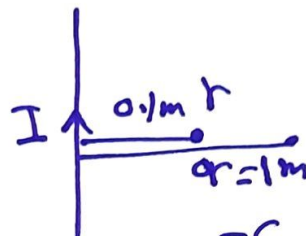
NOTE: $F_{\text{net}} = 0$.

(19.21) Given $N=50$, $I=2A$, $R=0.05m$

$$B = N \frac{\mu_0 I}{2R} = 50 \frac{4\pi \times 10^{-7} \times 2}{2 \times 0.05} = 1.3 \times 10^{-3} T$$

(19.25) Given $I=4A$

For a long wire $B = \frac{\mu_0 I}{2\pi r}$

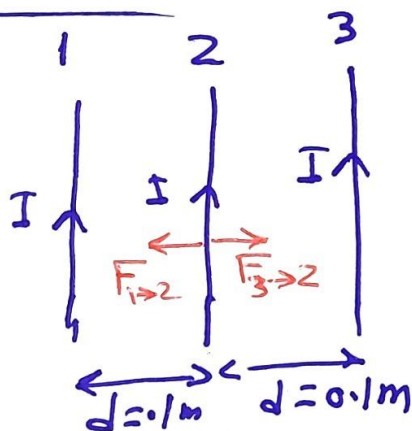


(a) $r=0.1m \Rightarrow B = \frac{4\pi \times 10^{-7} \times 4}{2\pi \times 0.1} = 8 \times 10^{-6} T$

(b) $r=1m \Rightarrow B = \frac{4\pi \times 10^{-7} \times 4}{2\pi \times 1} = 8 \times 10^{-7} T$

(19.30) $F_m = ILB \sin \theta$

(a) On Central wire
Since currents are parallel
 $\Rightarrow$ attractive forces



$$F_{1 \rightarrow 2} = F_{3 \rightarrow 2} = \frac{\mu_0 I_1 I_2}{2\pi d} L, \text{ since } I_1 = I_2$$

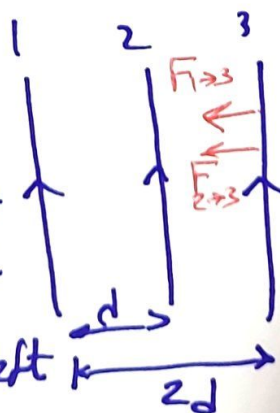
$$\vec{F}_{1 \rightarrow 2} + \vec{F}_{3 \rightarrow 2} = 0 \quad \text{equal in magnitude, opposite in direction}$$

(b) $F_{1 \rightarrow 3} = \frac{\mu_0 I^2}{2\pi(2d)} L$ (left)

$$F_{2 \rightarrow 3} = \frac{\mu_0 I^2}{2\pi d} L$$
 (left)

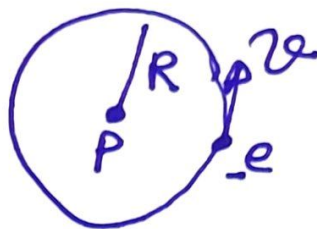
$$F = F_{1 \rightarrow 3} + F_{2 \rightarrow 3} = \frac{\mu_0 I^2 L}{4\pi d} + \frac{\mu_0 I^2 L}{2\pi d} = \frac{3\mu_0 I^2 L}{4\pi d}$$

$$= \frac{3 \times 4\pi \times 10^{-7} (10)^2}{4\pi \times 1} = 3 \times 10^{-4} N \text{ to the left}$$



(19.43) Given $R = 5.1 \times 10^{-11} \text{ m}$

$\omega = 6.8 \times 10^{15} \text{ Hz}$



(a) $I = \frac{q}{T} = q\omega$

$= 1.6 \times 10^{-19} \times 6.8 \times 10^{15} = 1.1 \times 10^{-3} \text{ A}$

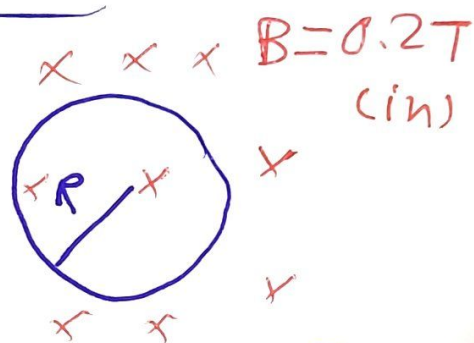
(b) $B = \frac{\mu_0 I}{2R} = \frac{4\pi \times 10^{-7} \times 1.1 \times 10^{-3}}{2 \times 5.1 \times 10^{-11}} = 13.5 \text{ T}$ (huge field)

(19.47) Given $R = 3 \text{ m}$

magnetic force $\rightarrow F_m = F_c \leftarrow \text{Centripetal force}$

$qvB \sin 90^\circ = m \frac{v^2}{R}$

$\Rightarrow v = \frac{qBR}{m} = \frac{1.6 \times 10^{-19} \times 0.2 \times 3}{1.6 \times 10^{-27}} = 6 \times 10^7 \text{ m/s}$



Q(1) ~~From~~ $m_1 = m_2 = m_3$

$v_1 = v_2 = v_3$

from question (19.47) above

$v = \frac{qBR}{m} \Rightarrow q = \left(\frac{mv}{B} \right) \frac{1}{R}$

$\frac{mv}{B} = \text{Constant say } \equiv k$

$q = \frac{k}{R} \Rightarrow q_1 > q_2 > q_3$ because $R_1 < R_2 < R_3$
radius of curvature

NOTE: q_3 is +ve charge, q_1, q_2 are -ve charges

